Internet Data Airlock – Solution Brief

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Data scientists need internet-sourced data, and the business needs a safe, auditable way to bring it into Corp. This brief covers an isolated quarantine zone, an airlock subscription and a one-way release into the Data Landing Zone (DLZ) and Fabric.

Business problems / drivers

- Data scientists need data scraped from public websites and APIs for analytics and AI use cases.
- Internet data is untrusted: it can carry malware, harmful content or personal information that must not reach the corporate environment.
- There is no approved path for internet data today, which pushes people to workarounds (laptop downloads, personal cloud storage, USB).
- Security requires internet-facing workloads to stay separated from Corp, the Hub and other spokes.
- Data must be classified before use to meet privacy and records obligations (Privacy Act / APPs).
- Every dataset entering Corp needs an auditable record of where it came from and who approved it.

Success criteria / requirements

- Data scientists get self-service, Data Science VM-style tooling with controlled internet access.
- The quarantine subscription has no network path to the Hub or any spoke, enforced by Azure Policy.
- Every file is malware-scanned; only files marked Clean move on, and files that can't be scanned are blocked.
- Every dataset is classified in Purview and given a sensitivity label before release.
- Release is gated: automatic when there are no findings, data steward approval for PII, Highly Confidential rejected or routed to a restricted area.
- Data moves one way only (quarantine → airlock → Corp), and Corp always initiates the pull.
- Full chain of custody per file: SHA-256 hash, scan result, classification, approver and timestamp.

- Released data lands in the DLZ and is available in Fabric through a OneLake shortcut.
- Everything is deployed as code (Bicep / AVM via Azure DevOps) and can be redeployed per use case.
- Targets: release SLA of [X hours, to agree] for clean, low-risk data; security alerts in Sentinel within 15 minutes.

User groups / personas

Persona	What they do	Zone they work in
Data scientist / researcher	Scrapes and prepares data on the VMs, submits it for release	Quarantine
Data steward / data owner	Reviews classification findings, approves or rejects releases	Airlock (approval workflow)
Data engineer	Builds the pipelines that pull released data into the DLZ and Fabric	Corp / DLZ
Platform / cloud engineering	Owns subscriptions, policy, network, IaC and day-to-day operations	All
Security operations (SOC)	Monitors Defender and Sentinel alerts, handles malware incidents	All (via Sentinel)
Privacy, risk and compliance	Sets classification rules, retention and the legal position on scraping	Governance
Data consumers (analysts, BI, AI teams)	Use the released data in Fabric	Corp / Fabric

Issues, constraints and assumptions

Assumptions

- One Entra tenant, with the Purview account in the Data Management Landing Zone (DMLZ).
- Region is Australia East and data must stay in Australia.
- Datasets are files (HTML, JSON, CSV, Parquet, PDF, images), not streaming data.
- No data needs to flow from Corp back into quarantine; if it does, a separate reverse airlock is needed.

Constraints

- Defender on-upload scanning handles up to 50 GB per minute per storage account; above that, files queue and some may not be scanned. Size and archive limits also apply, and encrypted or password-protected archives can't be scanned.
- Purview scans run on a schedule or by API trigger, not per file, so classification adds minutes to hours.
- Purview classifies and labels supported file types only; images and some binary formats get little or no classification.
- The airlock needs private endpoints from both sides. That isn't peering, but security must formally accept it.

Issues

- Scraping is subject to website terms of service and copyright, so legal sign-off is needed per source.
- Scraped content may include third-party personal information; a handling policy is needed (reject, mask or restrict).
- Data scientists will want tools and packages from the internet, so egress rules must balance flexibility and control.

Technical dependencies

Dependency	Needed for	Owner
Management group structure (new Quarantine MG, home for the airlock subscription)	Policy isolation	Platform
Subscription vending (quarantine + airlock subscriptions)	Landing the zones	Platform
Entra ID with PIM, Conditional Access and persona groups	Access control	Identity
Defender for Cloud: Servers P2, Storage with malware scanning	Malware and threat protection	Security
Purview account with managed VNet integration runtime and managed private endpoints	Classification	Data governance
Microsoft Sentinel + central Log Analytics workspace	Monitoring	SOC

Dependency	Needed for	Owner
DLZ with ADLS Gen2 and a Fabric capacity / workspace	Ingestion and consumption	Data platform
ADF or Fabric Data Factory with managed VNet	Corp-side pull from export	Data platform
Azure DevOps pipelines, AVM module library, workload-identity service connection per subscription	Deployment	Platform
Data Science VM marketplace images or an Azure Compute Gallery golden image	Data scientist workspace	Platform
Agreed classification tiers and sensitivity labels	Release gate rules	Data governance (open decision)

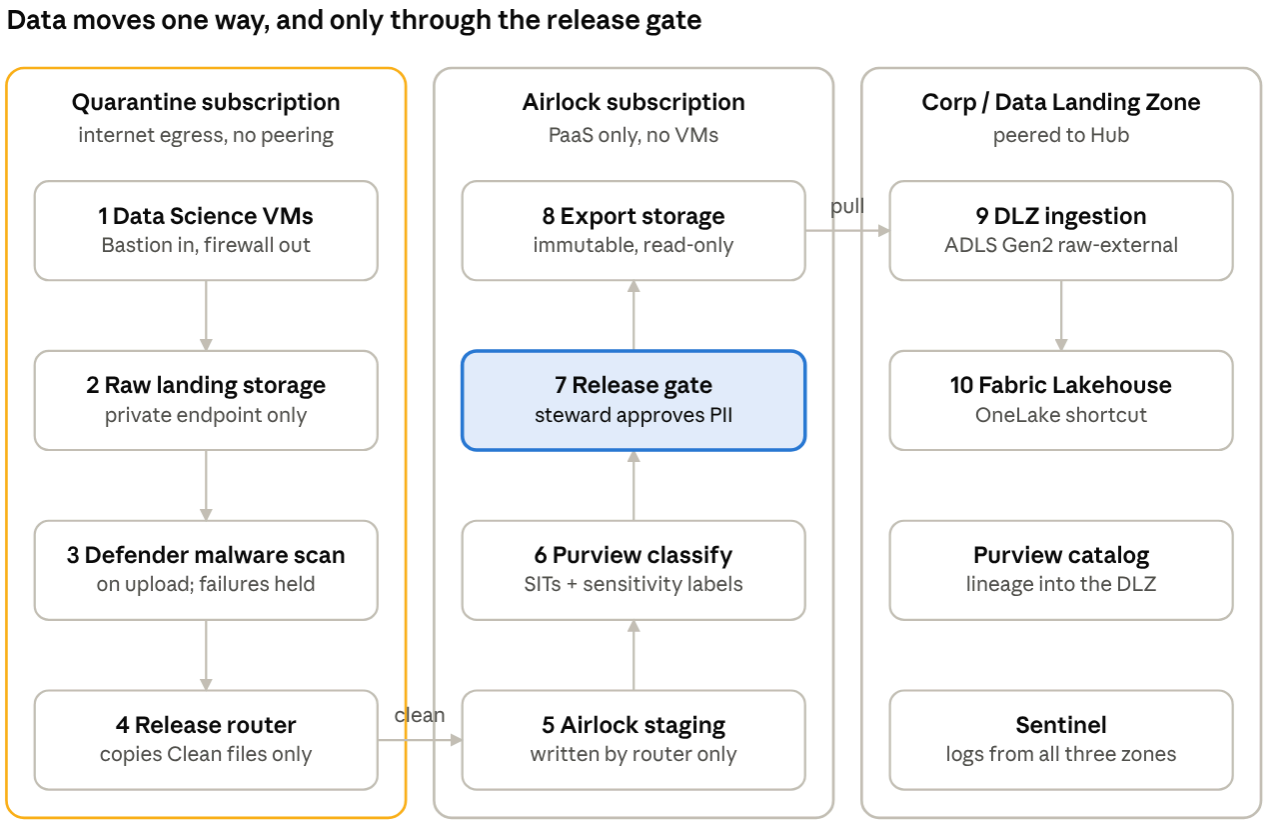
Architecture considerations

- **Isolation:** the quarantine VNet has no peering, gateways or Hub DNS; its private DNS zones are local to the subscription.
- **Egress:** a dedicated Azure Firewall Premium in quarantine, so internet traffic never touches the Hub firewall. Decide between a strict FQDN allow-list and broad web categories with IDPS.
- **One direction only:** quarantine can only write to airlock staging, Corp can only read from airlock export, and nothing writes into Corp.
- **Airlock as its own trust boundary:** a separate subscription that neither data scientists nor Corp admins control.
- **Event-driven with gates:** Defender scan result → Event Grid → Function → staging → Purview → Logic App gate → export.
- **Immutability:** WORM retention on export, versioning on staging, hashes re-checked when data lands in the DLZ.
- **Scale:** the 50 GB/min scan limit per account means very large volumes need several raw accounts or batching.
- **Compute:** Data Science VMs to start; Azure ML compute instances or AVD if user experience or manageability becomes a problem.
- **Log routing:** diagnostics go through Azure Monitor to central Log Analytics, which needs no network path between zones.

- **Repeatable pattern:** an AVM-based module set, so each new research group or project can get its own airlock.

Solution overview / components

Three zones, one direction: data is scraped and scanned in quarantine, classified and approved in the airlock, then pulled into Corp.



airlock architecture · 3 zones, 10 steps

Follow the numbers: down through quarantine, up through the airlock, then down into Corp. Nothing flows back the other way.

Quarantine subscription (internet zone)

- VNet with subnets for Bastion, VMs, firewall and private endpoints.
- Azure Bastion (Standard) with Entra ID sign-in.
- Data Science VMs (Ubuntu or Windows), Entra-joined, with Defender for Endpoint and auto-shutdown.
- Azure Firewall Premium with a firewall policy (FQDN rules, web categories, IDPS, TLS inspection).
- Raw landing storage (ADLS Gen2, private endpoint only, shared key off) with Defender on-upload malware scanning.

- Quarantine-hold container for failed files, with soft delete and a purge policy.
- Release router: Event Grid plus a VNet-integrated Function with a managed identity; copies Clean files only and writes a SHA-256 manifest.

Airlock subscription (handover zone, no VMs)

- Staging storage, writable only by the release router.
- Purview managed private endpoints into staging.
- Logic App release gate with a data steward approval workflow.
- Export storage: immutable (WORM), read-only to Corp.
- Manifest and approval records (Table storage or Cosmos DB).

Corp / Data Landing Zone

- ADF or Fabric pipeline pulling from export over a managed private endpoint.
- DLZ ingestion storage for external raw data.
- Fabric Lakehouse with a OneLake shortcut.
- Purview Unified Catalog, with lineage carried through.

Cross-cutting

- Azure Policy initiative for Quarantine and Airlock.
- Sentinel analytics rules and playbooks.
- Budgets and cost alerts.
- Bicep / AVM code with Azure DevOps CI/CD.

Delivery cost and timeframe

A pilot (5 VMs, under 1 TB a month) runs at roughly A$3k–5k a month and takes 10–14 weeks to deliver. Unit prices below are approximate US list prices, not confirmed for Australia East; validate them in the Azure Pricing Calculator before budgeting.

Indicative monthly run cost

Component	Approximate cost	Notes
Azure Firewall Premium	~US$1.75/hr (~US$1.3k/month) + per GB processed	Largest fixed cost; Standard is cheaper if TLS inspection and IDPS aren't required
Azure Bastion Standard	~US$200–300/month	

Component	Approximate cost	Notes
Data Science VMs	~US$100–150/month each (D8s-class, 8×5)	GPU sizes cost far more
Defender for Servers P2	~US$15 per VM per month	
Defender for Storage	Per storage account + malware scanning per GB scanned	Set a monthly scan cap per account (Microsoft Learn)
Microsoft Purview	Data Map and scanning charges	Classic capacity or new governed-asset model, depending on tenant licensing (billing FAQ)
Storage, Functions, Logic Apps, Event Grid, Log Analytics	Usually small	Grows with data volume

Delivery plan (one team, ~10–14 weeks)

Phase	Duration	Scope
1. Discovery and design sign-off	2 weeks	Security and privacy review, legal position on scraping
2. Foundations	3 weeks	MG, subscriptions, policy, network, firewall, Bastion, all in IaC
3. Quarantine workspace and scanning	2–3 weeks	VMs, golden image, Defender, scan-and-route pipeline
4. Airlock	2–3 weeks	Staging and export, Purview integration, Logic App gate, manifest
5. Corp ingestion and monitoring	1–2 weeks	DLZ and Fabric pull, Sentinel rules
6. Testing and handover	2 weeks	EICAR malware tests, isolation bypass attempts, penetration test, runbooks

Team: a cloud/platform engineer, a data engineer, ~0.5 security engineer, ~0.25 Purview / governance specialist, and an architect for design assurance.

Cyber security considerations

- **Threat model:** malware in scraped files, a compromised VM used to reach other networks, a path to the Hub opened by mistake, and tampering with the release gate.

- **Network:** policy denies peering, gateways and public IPs outside the firewall; firewall IDPS in deny mode, malware and phishing categories blocked, TLS inspection where allowed.
- **Endpoint:** Defender for Endpoint with attack surface reduction rules on the VMs; never domain-joined, no corporate credentials, rebuilt regularly from a golden image.
- **Identity:**
  - Separate persona groups with PIM eligibility.
  - Conditional Access requires MFA and a compliant device for Bastion.
  - Bastion clipboard and file transfer restricted.
  - Shared keys and SAS disabled everywhere.
  - Managed identities scoped to the minimum (read-only or write-only).
- **Data:**
  - Unscannable files are blocked, not passed.
  - Optional file-type allow-list and content disarm and reconstruction (CDR) for Office and PDF files.
  - Hashes verified at every handover; export storage immutable.
  - Rejected and malicious data purged on a set schedule.
- **Separation of duties:** data scientists can't approve their own releases, and platform admins can't override the gate without an alert firing.
- **Monitoring:** Sentinel rules for malware detections, peering or private endpoint approval attempts, unusual egress volume, gate overrides and policy non-compliance; playbooks isolate the VM and freeze the raw container.
- **Supply chain:** packages on quarantine VMs come straight from the internet. Packages for Corp use come only through the private PyPI mirror with vulnerability gating, never through the airlock.
- **Assurance:**
  - Pre-production penetration test focused on breaking isolation.
  - Defender CSPM attack-path analysis.
  - Mapping to ACSC Essential Eight and ISM controls, if used.
  - Regular access reviews.